Notes: Diamond and Rajan (JF, 2000)
A Theory of Bank Capital

Contents

1 Big picture

- **Question.** Why does bank capital structure matter, and why do banks combine fragile demand deposits with softer outside capital?
- **Core idea.** Banks create liquidity precisely because their deposits are fragile. Fragility disciplines the banker by making it impossible for him to renegotiate depositors down after he has used their money to acquire relationship-specific collection skills.
- **Central trade-off.** Demand deposits maximize liquidity creation and credit creation because they commit the banker to pass through collections. But they also make the bank vulnerable to runs when asset values fall. Capital softens the liability side and provides a buffer, but it also reduces liquidity creation because capital holders can renegotiate.
- **Asset-side and liability-side integration.** The paper's main theme is that a bank's liability structure cannot be studied separately from what the bank does on the asset side. Bank liabilities shape the banker's bargaining power against borrowers and claimants.
- **Why banks are special.** An entrepreneur's project is illiquid because the entrepreneur's human capital cannot be pledged. A relationship lender learns how to liquidate or collect from the borrower, but the relationship lender's own human capital also cannot be pledged. A bank solves this second problem by issuing demand deposits.
- **Why demand deposits are useful.** Demand deposits create a sequential-service run threat. If the banker tries to hold up depositors by withholding collection skills, depositors run, disintermediate the banker, and eliminate his rent.
- **Why bank capital is costly.** Capital is a softer claim. It can be renegotiated, so capital holders cannot credibly commit to punish the banker in the same immediate way as depositors. More capital therefore increases banker rents and reduces pledgeability.
- **Why bank capital is beneficial.** Capital absorbs shocks and reduces the probability that nonverifiable declines in asset value trigger inefficient runs.
- **Multiperiod insight.** Bank leverage also changes how much the bank can extract from borrowers. A highly levered bank may have a shorter horizon and a stronger immediate liquidation threat, which can increase repayment extraction from cash-constrained borrowers.
- **Policy takeaway.** Capital requirements and deposit insurance affect not only bank safety but also liquidity creation, borrower bargaining, credit supply, and the distribution of rents between banks and firms.

2 Incremental contribution of the paper

2.1 Contribution relative to standard corporate capital-structure theories

Most work on capital structure before this paper treated bank capital by analogy with industrial-firm capital structure. Diamond and Rajan instead build a theory in which bank capital is endogenous to the bank's special role as a relationship lender and liquidity creator. The paper therefore moves the question from "why do banks hold equity?" to "how does bank capital change the bank's ability to create liquidity, discipline bankers, and extract loan repayment?"

Incremental contribution:

- It gives a banking-specific reason why capital structure matters even without standard asymmetric-information issuance costs.
- It links capital structure to the bank's asset-side collection technology, not just to liability-side risk sharing.

- It shows that bank capital is not simply a buffer against insolvency; it changes the banker's ability to pledge loan collections to outsiders.

2.2 Contribution to the theory of demand deposits and bank fragility

The paper turns bank fragility from a defect into a productive commitment mechanism. Fragile deposits discipline the banker because depositors' ability to run prevents renegotiation.

Incremental contribution:

- It provides a microfoundation for why banks use demandable debt: demandability creates a collective-action problem that commits depositors not to make concessions.
- It explains why the same feature that creates liquidity also creates fragility.
- It clarifies why deposit insurance can weaken the disciplinary role of deposits if it removes depositors' incentive to run.

2.3 Contribution to incomplete-contracts and relationship-lending theory

The paper uses the inalienability of human capital to explain both real-asset illiquidity and financial-asset illiquidity. Entrepreneurs cannot pledge their specific skills; bankers cannot pledge their specific collection skills.

Incremental contribution:

- It applies incomplete-contracts logic to the inside of banks, not only to firms and workers.
- It shows that the bank's organizational form can make relationship-lending skills pledgeable.
- It explains why relationship lenders need a special liability structure to finance themselves.

2.4 Contribution to regulatory-capital debates

Capital is usually treated as unambiguously stabilizing. This paper highlights that capital has both stabilizing and credit-reducing effects.

Incremental contribution:

- Capital requirements can reduce liquidity creation and credit supply.
- Abrupt increases in future capital requirements can shorten the banker's horizon and change borrower repayment incentives.
- Recapitalization policy affects the distribution of bargaining power between banks and borrowers, not only the probability of failure.

2.5 Contribution to historical interpretation

The paper provides a theory for the long-run decline in bank capital ratios. As markets improve and project assets become more liquid, banks need less capital because the cost of financial distress falls relative to the liquidity-creation benefits of deposits.

Incremental contribution:

- It explains falling bank capital without relying only on deposit insurance or regulation.
- It links financial development to more deposit-intensive bank balance sheets.
- It predicts matching between bank capital structures and borrower types.

3 Model objects and measurement

3.1 Agents and assets

- There are entrepreneurs with projects, investors, and later a banker/intermediary.
- A project requires one unit of investment and returns cash flow C^s in state $s \in \{H, L\}$.
- Cash flow is produced only if the entrepreneur contributes human capital.
- The project's assets have a liquidation value X^s before cash flow is produced.
- The entrepreneur cannot commit future human capital, so a financier can collect only by threatening liquidation.

Detailed interpretation: The project is illiquid because the entrepreneur is the best user of the asset. The liquidation value is lower than the value under the entrepreneur's human capital.

3.2 Relationship lender and specific collection skills

A lender who finances the project early learns how to redeploy the project assets. This creates relationship-specific collection ability.

- The initial lender can liquidate for X^s.
- A replacement lender can liquidate for only βX^s, where $\beta < 1$.
- The lender's relationship-specific skill raises pledgeability for the entrepreneur.
- But the lender's skill also creates a new hold-up problem: the lender may be able to hold up outside investors who financed the loan.

Detailed interpretation: The same friction appears twice. The entrepreneur's human capital makes the real asset illiquid. The banker/lender's human capital makes the loan illiquid unless the lender is disciplined.

3.3 Demand deposits

Demand deposits are fixed claims with a sequential-service feature. Depositors can demand repayment and are paid in order until cash/assets run out.

- If the banker behaves, deposits are repaid from collections.
- If the banker threatens to withhold collection skills, depositors can run.
- A run disintermediates the banker and eliminates his rent.
- This threat commits the banker to pass through collections.

Detailed interpretation: Deposits are useful because they are fragile. Their fragility prevents banker renegotiation.

3.4 Capital

Capital is a long-term, non-demandable claim. It is most naturally interpreted as outside equity, but can also represent subordinated debt with control rights after default.

- Capital holders can replace or liquidate the banker.

- Capital holders are not subject to the immediate collective-action problem that makes depositors run.
- Capital can renegotiate with the banker.
- Therefore capital buffers shocks but weakens the banker's commitment.

Detailed interpretation: Capital is soft. It provides insurance against bad states but leaves rents to the banker because capital holders cannot commit to an immediate run.

3.5 Safe versus risky deposit structures

The paper compares two liability structures:

- **Safe structure:** deposits are low enough that no run occurs even in the low state.
- **Risky structure:** deposits are high enough to eliminate banker rents in the high state, but trigger a run in the low state.

Detailed interpretation: The optimal structure depends on whether the expected gain from eliminating banker rents exceeds the expected cost of a run.

4 Models and theoretical specifications

4.1 Entrepreneur-lender bargaining

At date k, the entrepreneur offers current and future payments. If the lender accepts, the entrepreneur contributes human capital and payments are made. If the lender rejects, the lender can liquidate or renegotiate.

$$\text{Outside lender liquidation value} = \beta^j X_k, \tag{1}$$

where $\beta^j = 1$ for the initial lender and $\beta^j = \beta < 1$ for other lenders.

Detailed interpretation: The initial lender's advantage comes from relationship-specific knowledge. This makes relationship lending valuable but creates nonpledgeable rents.

4.2 Banker hold-up of capital

If the banker has borrowed from capital holders, he can threaten not to collect the loan unless capital makes concessions. Figure 2 formalizes this bargaining.

$$\text{Capital's fallback value} = \max\{\beta X^s - d, 0\}, \tag{2}$$

where d is promised deposit repayment.

Detailed interpretation: Capital needs the banker to collect more than an arm's-length liquidation value. Since capital cannot credibly run, the banker can extract part of the surplus from capital.

4.3 Depositor run discipline

When the banker threatens depositors, depositors can run and seize bank assets sequentially. Running prevents the banker from extracting rents because the banker is no longer necessary after depositors seize the claims.

$$\text{Run payoff depends on sequential priority and the market value of seized loans.} \tag{3}$$

Detailed interpretation: Depositors are strong claimants not because they negotiate well, but because they cannot coordinate to renegotiate. Their collective-action problem is a commitment device.

4.4 Lemma 1: payments with deposits and capital

Lemma 1 characterizes outcomes depending on promised payments P_2, deposits d, and liquidation values.

- If the bank cannot cover deposits, depositors run.
- If the bank can cover deposits but capital needs the banker to collect, capital and banker split surplus.
- If capital can collect enough without the banker, the banker earns little or no rent.

Detailed interpretation: The lemma decomposes how value is divided among depositors, capital, and banker. Depositors get fixed priority; capital and banker bargain over residual surplus.

4.5 Safe and risky financing amounts

When deposits are low enough to be safe, expected pledgeable value is denoted D^{Safe}. When deposits are high enough to create a run in the low state but eliminate banker rent in the high state, pledgeable value is denoted D^{Risky}.

One key benchmark case is:

$$D^{Safe} \approx q_H \frac{1+\beta}{2} X^H + (1-q_H) X^L, \tag{4}$$

$$D^{Risky} \approx q_H X^H + (1-q_H) \frac{1+\beta}{2} X^L. \tag{5}$$

Detailed interpretation: The risky structure gets more in the high state because deposits eliminate banker rents, but loses value in the low state because a run destroys the banker's collection value.

4.6 Lemma 2: when safe or risky structure raises more

Lemma 2 compares D^{Safe} and D^{Risky}:

- If low-state losses from distress are large relative to high-state rents, $D^{Safe} > D^{Risky}$.
- If high-state rents are large relative to low-state distress costs, $D^{Risky} > D^{Safe}$.
- For intermediate cases, a threshold in project liquidity β determines which structure raises more.

Detailed interpretation: The optimal deposit level is a leading indicator of expected conditions: safer capital structures are preferred when bad states are important; deposit-heavy structures are preferred when high-state rent extraction is important.

4.7 Corollary 1: risk and leverage

The paper shows that under the relevant conditions, $D^{Risky} - D^{Safe}$ increases with a mean-preserving spread in X.

Detailed interpretation: Riskier loan repayments can be associated with more deposit leverage, not because the bank is shifting risk to depositors, but because value in high states makes commitment to pay out collections more important.

4.8 Multiperiod projects and borrower extraction

In a multiperiod setting, deposits maturing at date 1 affect the banker's horizon and bargaining stance toward the borrower.

Capital's expected payment after rejecting the banker can be written schematically as:

$$\pi_1^s = \frac{1}{2}\max\{\beta \max(X_1^s, E[X_2|s]) - d_1, 0\} + \frac{1}{2}\max\{X_1^s - d_1, 0\}. \tag{6}$$

The bank accepts the borrower's payment offer only if:

$$P_1^s + \text{Pledgeable}(P_2^s) \geq \pi_1^s + d_1. \tag{7}$$

Detailed interpretation: Higher maturing deposits can force the banker to demand more current payment or liquidate. Bank capital structure therefore affects borrower bargaining and credit crunch dynamics.

5 Theoretical design and identification logic

5.1 Why the model is not a standard moral-hazard model

The banker does not create value by exerting costly unobservable effort. The banker's specific collection skill is valuable because it affects bargaining and liquidation outcomes. The bank is an institutional device for tying this specific human capital to loans.

5.2 Why demand deposits are not just short-term debt

Short-term debt can mature, but if creditors are treated symmetrically they may renegotiate. Demand deposits create priority through sequential service. Some claimants can get priority only by demanding payment immediately, which creates a run threat.

5.3 Why entrepreneurs cannot issue deposits directly

The entrepreneur is still necessary after depositors seize the assets. Depositors would rehire him and pay his rents. The banker, by contrast, is redundant after depositors seize the loan claim; this is why demand deposits discipline a bank but not the underlying borrower.

5.4 What the paper cannot prove empirically

- The paper is theoretical and does not estimate model parameters.
- The model abstracts from taxes, deposit insurance pricing, agency conflicts among multiple bank managers, and regulatory resolution procedures.

- The bargaining protocol is stylized and chosen for tractability.
- The model explains mechanisms rather than delivering a quantitative capital-ratio target.

Detailed interpretation: The theoretical design isolates the balance-sheet mechanism: capital changes liquidity creation because it changes the ability of bank claimants to commit against renegotiation.

6 Section-by-section study guide

6.1 Introduction

Focus on the framing: bank capital affects liquidity creation, distress costs, and borrower extraction. The key departure is that banks are not treated like ordinary firms.

6.2 Section I: One-period framework

This section builds the core mechanism. The entrepreneur cannot pledge human capital. The lender learns specific collection skills. The bank finances the lender with claims that either discipline or soften the lender.

6.3 Figures 1–3

The three figures are bargaining trees. Figure 1 describes entrepreneur-lender bargaining. Figure 2 adds a banker financed by capital. Figure 3 replaces capital with demand deposits and shows why runs discipline the banker.

6.4 Lemma 1 and Lemma 2

Lemma 1 gives the payoff division with deposits and capital. Lemma 2 gives the safe-versus-risky capital-structure comparison. Read them as the paper's main computational engine.

6.5 Section II: Multiperiod projects

This section adds borrower cash timing. Bank capital now affects how much the bank extracts and whether it liquidates. The bank's balance sheet changes its bargaining horizon.

6.6 Section III: Implications

The paper applies the mechanism to capital requirements, deposit insurance, bank capital crises, borrower matching, and financial development.

6.7 Section IV: Robustness

The paper explains why alternative claims, entrepreneur-issued deposits, cash, and state-contingent contracts do not overturn the core mechanism.

7 Tables and figures

This paper has three theoretical figures and no empirical tables. The figures are directly extracted from the paper. Close figures are placed on the same row; the remaining single figure uses width 0.6\linewidth as requested.

7.1 Figure 1: Bargaining between Entrepreneur and Lender + Figure 2: Bargaining within an Intermediary

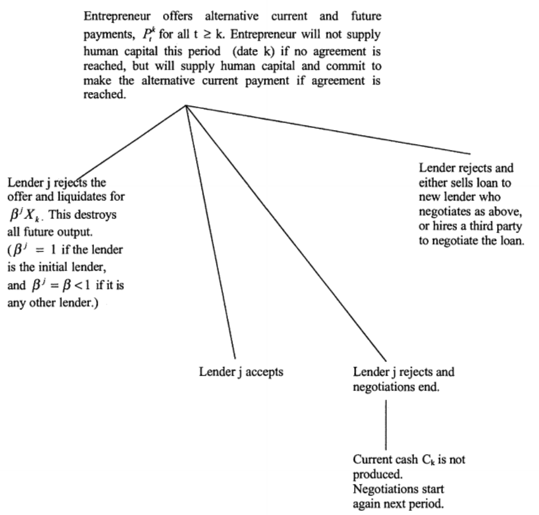

Figure 1. Bargaining between entrepreneur and lender *j* at date *k*.

(a) Figure 1: Bargaining between entrepreneur and lender j at date k

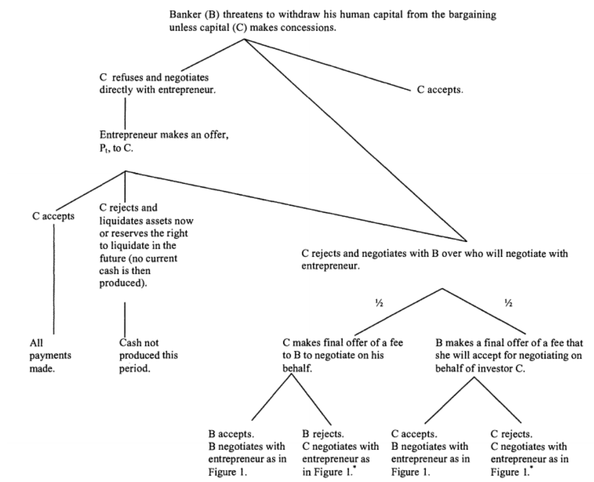

Figure 2. Bargaining within an intermediary. *If this part of the tree is entered after *C* rejects the entrepreneur's offer, *C* now has the option of liquidating or negotiating with the entrepreneur but not of accepting the initial offer.

(b) Figure 2: Bargaining within an intermediary

Read:

- Figure 1 starts from the entrepreneur's offer of current and future payments. The lender can accept, reject and liquidate, reject and end negotiations, or transfer/sell the loan to another negotiator.
- The figure emphasizes that the entrepreneur's human capital is supplied only after agreement. Without agreement, cash is not produced.
- The original lender has a stronger liquidation threat than a replacement lender: $\beta^j = 1$ for the original lender but $\beta^j = \beta < 1$ for another lender.
- Figure 2 embeds the lender inside an intermediary. The banker can threaten capital holders by withholding collection skills unless capital makes concessions.
- Capital can accept, negotiate directly with the entrepreneur, or bargain over who negotiates with the entrepreneur.

Detailed interpretation: Figure 1 shows why lending is relationship-specific: the initial lender has learned how to collect and liquidate. Figure 2 shows the central financing problem of relationship lending: once the banker has acquired collection skills, he can hold up the investors who funded him. The banker's knowledge makes the loan more valuable but also makes the loan hard to pledge.

Economic message: Relationship lending creates value, but the value is trapped in the relationship lender's human capital. A bank is useful only if its liabilities can credibly tie that human capital to outside investors.

7.2 Figure 3: Bargaining with Depositors

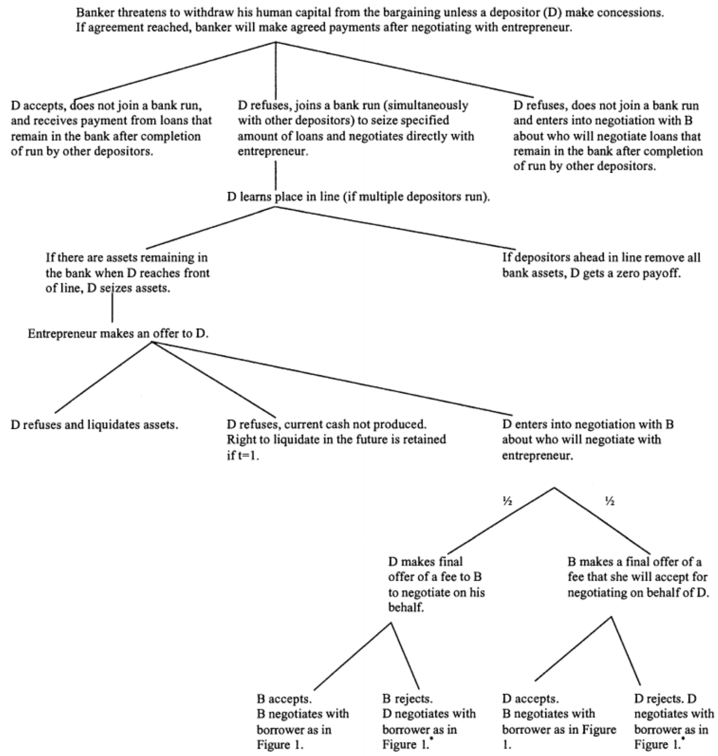

Figure 3. Bargaining with depositors. *If this part of the tree is entered after D rejects the entrepreneur's offer, D now has the option of liquidating or negotiating with the entrepreneur but not of accepting the initial offer.

Figure 3: Bargaining with depositors

Read:

- The banker threatens to withhold collection skills unless an individual depositor makes concessions.
- The depositor can accept and stay, or refuse and join a run with other depositors.
- If the depositor joins a run, his payoff depends on his place in the sequential-service line.
- If assets remain when he reaches the front of the line, he seizes assets; if earlier depositors exhaust assets, he receives zero.
- The collective-action problem makes depositors unwilling to wait and renegotiate.

Detailed interpretation: Figure 3 is the paper's most important intuition. Depositors are weak individually but strong collectively because they cannot coordinate to give concessions. Their individual incentive to run makes the banker's threat to withhold skills ineffective. The run threat disciplines the banker and forces him to pass collections through.

Economic message: Bank fragility creates liquidity. The same run mechanism that can destroy value in bad states is what allows the bank to pledge more loan value in normal states.

8 Detailed result map

8.1 Result block 1: Why projects and loans are illiquid

Result block: The entrepreneur cannot pledge human capital; the relationship lender cannot pledge collection human capital.

- The project is illiquid because the entrepreneur is the best user of the assets.
- The loan is illiquid because the banker is the best collector of the loan.
- The bank's problem is to make the banker's human capital pledgeable.

Economic message: Illiquidity comes from nontransferable human capital, not merely from asymmetric information or transaction costs.

8.2 Result block 2: Why deposits are optimal under certainty

Result block: In a world of certainty, all-deposit financing maximizes liquidity creation.

- Deposits create a run threat.
- The run threat prevents banker rent extraction.
- Lower banker rent means more can be raised up front.

Economic message: In the certainty case, fragility is purely productive because it disciplines the banker without causing inefficient runs.

8.3 Result block 3: Why capital becomes useful under uncertainty

Result block: With noncontractible uncertainty, risky deposits can cause inefficient runs; capital can absorb shocks.

- High deposits maximize commitment in high states.
- But high deposits trigger runs in low states.
- Capital reduces run risk but allows banker renegotiation.

Economic message: Capital is useful because it reduces fragility, but costly because it weakens commitment.

8.4 Result block 4: Why bank capital affects borrowers

Result block: In multiperiod projects, bank leverage changes the banker's bargaining horizon and affects borrower payments/liquidation.

- A high deposit obligation can make the banker prefer immediate liquidation.
- A better capitalized bank can wait for future repayments.
- Borrowers with different cash-flow timing may match with banks of different capital structures.

Economic message: Bank capital is not only about bank safety; it changes the terms of credit for borrowers.

8.5 Result block 5: Policy implications

Result block: Capital requirements and deposit insurance have subtle effects because they alter commitment, not only solvency.

- Higher capital requirements can reduce credit and liquidity creation.
- Abrupt future capital requirements can shorten bank horizons and intensify credit crunches.
- Deposit insurance can weaken run discipline if it fully removes depositors' incentives to run.

Economic message: Safety regulation can reduce the very fragility that makes banks useful.

9 Short summary

- The paper develops a theory of bank capital rooted in the bank's asset-side function as a relationship lender.
- Entrepreneurs cannot pledge human capital, so projects are illiquid.
- Relationship lenders learn how to collect more from borrowers, but their own collection skills are also nonpledgeable.
- Demand deposits make banker skills pledgeable by creating a run threat.
- Demand deposits are therefore useful because they are fragile.
- Capital is a soft claim that absorbs shocks but lets the banker bargain for rents.
- Optimal bank capital trades off liquidity creation, distress costs, and borrower extraction.
- In a one-period model, the bank chooses between safe deposits and risky deposits depending on high-state rent extraction versus low-state run costs.
- In a multiperiod model, deposits affect the banker's horizon and hence the amount extracted from borrowers.
- Capital requirements can reduce liquidity creation and can affect borrower payments.
- Deposit insurance can reduce the disciplinary role of deposits.
- The theory helps explain the decline in bank capital ratios as financial development improves asset liquidity.

10 Conclusion

10.1 Core conclusion

Diamond and Rajan provide a theory in which bank capital structure matters because bank liabilities discipline the banker's human capital. Banks create liquidity by financing relationship loans with fragile deposits. Capital makes banks safer, but it reduces the commitment value of deposits and therefore reduces liquidity creation.

10.2 Economic intuition

The core intuition is that a banker's collection skill is valuable but hard to pledge. If the banker could freely renegotiate with investors, he would capture part of the loan value as a rent. Demand deposits solve this by making investors unable to coordinate. If the banker threatens them, they run. This removes the banker and destroys his rent. The run threat makes his skill pledgeable.

10.3 Why bank capital is not ordinary corporate equity

In an industrial firm, equity is a residual claim that absorbs risk. In this banking model, capital also changes the bank's ability to commit. Since capital holders can negotiate with the banker, capital softens the bank's liability side. This softening is valuable in bad states but costly in normal states.

10.4 Regulatory implication

Capital requirements improve safety but can reduce the bank's credit-creation capacity. A regulator who sees only solvency may overstate the benefits of capital and understate the cost in lost liquidity creation. Conversely, a system with too few capital buffers is prone to inefficient runs. The theory therefore supports a trade-off view of prudential regulation rather than a one-sided safety view.

10.5 Deposit insurance implication

Deposit insurance can stabilize banks, but if it fully removes depositor run incentives, it also removes the commitment device that makes deposits useful. The model therefore suggests that insurance, closure rules, and supervisory intervention must preserve some form of discipline if banks are to continue creating liquidity effectively.

10.6 Borrower implication

Bank capital structure affects borrowers because it changes the banker's threat point. A highly levered bank may liquidate or extract more from cash-poor borrowers. A well-capitalized bank may be more patient and better suited to borrowers whose value arrives later. Thus bank capital can shape borrower-bank matching.

10.7 Historical implication

The paper explains why bank capital ratios may decline as financial markets develop. When projects and loans become more liquid, the cost of runs falls and the value of deposit discipline rises relative to capital buffers. More developed financial systems can therefore support more deposit-heavy bank balance sheets.

10.8 Limitations

The paper is not a quantitative calibration. It abstracts from taxes, explicit supervisory rules, heterogeneity among depositors, bank competition, securitization, and modern resolution regimes. It also relies on stylized bargaining protocols. These simplifications are intentional because the goal is to isolate the connection between fragile liabilities and liquidity creation.

10.9 Takeaway

The paper's enduring insight is that bank fragility is not merely a pathology. In the right institutional setting, fragility is what makes banks useful. Bank capital is therefore a double-edged instrument: it stabilizes banks, but it can also weaken the mechanism through which banks create liquidity and discipline relationship lenders.